

MODULE 12

SQL Stored Procedures and Functions

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

SQL Stored Procedures and Functions

Stored Procedures

Saved SQL code executed repeatedly.

Benefits: reduce network traffic,
improve performance, enforce business logic.

Create Procedure

```
DELIMITER //
CREATE PROCEDURE GetStudents(IN dept VARCHAR(50))
BEGIN
SELECT first_name, gpa FROM students
WHERE department = dept ORDER BY gpa DESC;
END //
DELIMITER ;

CALL GetStudents('Computer Science');
```

OUT Parameters

```
DELIMITER //
CREATE PROCEDURE DeptStats(
IN dept VARCHAR(50), OUT avg_gpa DECIMAL(3,2)
)
BEGIN
SELECT AVG(gpa) INTO avg_gpa
FROM students WHERE department = dept;
END //
DELIMITER ;

CALL DeptStats('CS', @avg); SELECT @avg;
```

User Functions

```
DELIMITER //
CREATE FUNCTION Grade(gpa DECIMAL(3,2))
RETURNS VARCHAR(2) DETERMINISTIC
BEGIN
IF gpa >= 3.7 THEN RETURN 'A+';
ELSEIF gpa >= 3.0 THEN RETURN 'B+';
ELSE RETURN 'B';
END IF;
END //
DELIMITER ;

SELECT first_name, Grade(gpa) FROM students;
```

Control Flow

```
IF cond THEN ... ELSEIF cond THEN ... ELSE ... END IF;
```

```
WHILE i <= 10 DO  
SET total = total + i;  
SET i = i + 1;  
END WHILE;
```

```
CASE x WHEN 'A' THEN ... ELSE ... END CASE;
```